

LIMON, CO, USA

WMO: 724665

Lat: 39.275N

Lon: 103.666W

Elev: 1638

StdP: 83.13

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 93010

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-20.6	-17.1	-23.6	0.5	-17.6	-20.7	0.7	-12.2	15.1	-3.1	14.0	-3.0	2.7	350	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	34.1	16.0	32.7	16.0	31.3	15.8	19.0	27.2	18.3	26.9	17.7	26.5	5.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	14.7	20.5	16.1	14.0	20.1	15.1	13.1	19.7	60.6	27.1	58.2	26.7	56.0	26.2	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.2	11.5	9.9	DB	-25.9	36.6	3.0	1.4	-28.0	37.6	-29.8	38.4	-31.5	39.1	-33.7	40.1	(4)
(5)			WB	-26.3	20.5	3.0	0.8	-28.5	21.1	-30.2	21.6	-31.9	22.0	-34.1	22.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	-2.7	-2.0	3.0	7.0	12.5	18.4	21.8	20.5	16.0	8.5	2.1	-3.1	(6)	
	DBStd	9.97	5.63	5.83	5.26	4.73	4.38	3.41	2.26	2.45	3.86	4.97	5.38	5.77	(7)	
	HDD10.0	1818	392	337	219	109	27	1	0	0	6	84	238	407	(8)	
	HDD18.3	3813	651	569	474	339	185	39	4	9	87	304	486	665	(9)	
	CDD10.0	1298	0	1	4	20	103	254	366	326	184	38	2	0	(10)	
	CDD18.3	252	0	0	0	0	4	43	112	77	16	0	0	0	(11)	
	CDH23.3	4883	0	1	3	32	231	948	1767	1209	599	91	3	0	(12)	
	CDH26.7	2088	0	0	0	2	58	406	886	509	212	14	1	0	(13)	
(14) Wind	WSAvg	4.2	3.7	4.2	4.7	5.4	4.8	4.4	3.9	3.8	3.8	4.1	3.8	3.8	(14)	
(15) Precipitation	PrecAvg	412	9	9	19	36	57	62	74	72	29	24	13	6	(15)	
	PrecMax	512	23	27	49	134	115	120	180	104	79	61	36	18	(16)	
	PrecMin	341	0	0	3	3	13	27	15	33	6	0	1	0	(17)	
	PrecStd	49	7	8	14	27	34	31	39	25	20	21	11	6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.1	19.7	23.5	26.4	30.8	35.7	36.4	34.7	32.8	28.4	23.4	17.4	(19)	
		MCWB	6.1	6.7	8.5	10.0	13.9	14.7	16.7	16.1	15.0	12.0	8.4	6.0	(20)	
	2%	DB	14.3	15.9	21.0	23.9	28.0	33.0	34.7	32.9	31.1	26.1	20.6	13.9	(21)	
		MCWB	4.2	5.0	7.5	9.6	12.9	15.2	16.2	16.1	14.3	11.2	7.5	4.2	(22)	
	5%	DB	11.1	12.9	18.3	21.9	26.1	31.2	33.2	31.5	29.1	23.4	17.3	11.0	(23)	
		MCWB	3.1	3.9	6.6	8.8	12.5	15.0	16.4	16.2	14.0	10.2	6.6	3.0	(24)	
	10%	DB	7.6	9.1	15.2	18.9	23.5	29.0	32.0	29.8	27.3	20.3	13.6	7.5	(25)	
		MCWB	1.5	2.2	5.4	8.1	12.0	15.1	16.5	16.2	13.8	9.3	4.9	1.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	7.1	9.7	12.1	16.6	19.2	20.1	19.9	17.7	14.0	9.5	6.4	(27)	
		MCDB	17.0	17.8	20.5	20.9	26.0	26.7	27.9	27.6	26.1	22.8	20.2	16.6	(28)	
	2%	WB	4.7	5.5	8.3	10.7	15.0	18.0	19.1	18.8	16.5	12.5	8.0	4.6	(29)	
		MCDB	13.4	15.0	18.8	20.5	23.3	26.3	27.5	27.1	24.9	21.8	18.8	13.2	(30)	
	5%	WB	3.3	4.1	7.1	9.7	13.9	17.1	18.5	18.0	15.7	11.3	6.9	3.0	(31)	
		MCDB	10.6	12.1	17.1	19.6	22.2	25.8	27.2	26.4	24.2	20.5	16.8	10.3	(32)	
	10%	WB	1.6	2.5	5.7	8.7	12.8	16.2	17.9	17.5	14.9	10.0	5.2	1.4	(33)	
		MCDB	7.4	9.0	14.5	17.8	21.1	25.2	27.0	25.7	23.4	19.0	13.2	7.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	17.4	16.6	17.9	16.9	16.6	17.5	18.1	16.9	18.4	18.1	18.3	17.2	(35)	
		MCDBR	22.3	22.8	23.7	23.8	21.8	22.0	21.8	20.4	22.5	24.1	24.3	22.5	(36)	
	5% WB	MCWBR	13.4	13.1	12.1	11.1	8.9	7.6	6.9	6.6	8.5	11.2	13.0	13.7	(37)	
		MCDWR	21.3	21.2	21.6	20.8	18.2	18.0	17.1	16.5	18.4	20.0	23.1	21.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.235	0.243	0.265	0.292	0.316	0.329	0.352	0.343	0.306	0.267	0.248	0.237	(40)	
		taud	2.614	2.607	2.541	2.476	2.469	2.450	2.435	2.451	2.523	2.620	2.589	2.591	(41)	
	Ebn at Noon		975	1006	1004	986	961	944	919	922	943	962	946	945	(42)	
		Edh at Noon	67	78	93	106	109	112	113	108	94	76	68	63	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.60	3.47	4.76	5.86	6.66	7.49	7.34	6.56	5.43	3.97	2.86	2.27	(44)	
	RadStd		0.14	0.22	0.34	0.33	0.45	0.32	0.28	0.19	0.26	0.26	0.18	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+70	N/A

Nomenclature: See separate page